

BITOLA, North Macedonia

WMO: 135830

Lat: 41.050N

Lon: 21.367E

Elev: 589

StdP: 94.45

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-9.2	-16.0	1.0	-12.7	-12.1	1.4	-6.5	8.1	8.8	7.2	8.2	0.5	0	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	34.0	18.9	32.3	18.5	30.8	18.1	20.2	30.4	19.5	29.3	18.8	28.2	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.2	21.8	16.5	12.6	21.3	15.8	12.0	20.9	60.4	30.5	58.0	29.1	55.8	28.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.3	5.2	-17.4	36.5	5.2	1.6	-21.1	37.7	-24.2	38.6	-27.1	39.5	-31.0	40.7		
(5)			-17.0	21.5	4.9	0.9	-20.6	22.2	-23.4	22.7	-26.2	23.3	-29.8	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	0.1	3.3	7.6	11.6	16.0	19.9	22.7	22.7	18.0	12.7	7.5	2.0		
	DBStd	8.57	5.64	4.73	3.78	3.61	2.92	3.05	2.66	2.37	3.28	3.81	4.03	5.02		
	HDD10.0	977	307	191	93	22	1	0	0	0	0	21	94	250		
	HDD18.3	2678	564	422	334	203	84	17	3	1	44	176	324	505		
	CDD10.0	1728	1	2	18	69	185	298	392	395	241	104	19	3		
	CDD18.3	388	0	0	0	1	11	66	137	138	35	1	0	0		
	CDH23.3	4679	0	0	0	23	167	813	1643	1578	413	41	0	0		
	CDH26.7	1848	0	0	0	2	29	288	718	684	123	4	0	0		
(14) Wind	WSAvg	1.7	1.4	1.8	2.2	2.2	1.8	1.7	1.8	1.6	1.6	1.4	1.5	1.4		
(15) Precipitation	PrecAvg	603	49	42	45	57	55	35	31	33	57	67	63	70		
	PrecMax	826	118	88	87	118	112	93	97	59	164	149	138	169		
	PrecMin	386	2	12	10	15	12	5	2	6	4	0	17	0		
	PrecStd	114	29	23	22	28	25	22	24	15	40	41	31	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.4	17.5	21.7	26.1	29.7	34.2	36.6	35.9	32.3	26.6	21.2	17.1			
	0.4% MCWB	8.0	9.6	11.0	13.7	17.2	18.5	19.2	19.5	17.5	15.5	12.5	10.6			
	2% DB	11.2	14.5	18.9	22.8	27.3	31.8	34.0	34.0	30.1	24.3	18.4	13.4			
	2% MCWB	6.9	8.0	9.8	12.8	16.0	18.2	19.1	19.0	17.0	14.3	11.7	9.0			
	5% DB	9.3	12.2	16.9	20.7	25.2	30.0	32.4	32.2	28.0	22.1	16.1	11.0			
	5% MCWB	6.0	7.1	9.0	11.8	15.2	18.1	18.6	18.6	16.2	13.5	10.7	8.1			
	10% DB	7.4	10.2	14.7	18.6	23.4	28.0	30.8	30.6	25.9	20.0	14.1	9.2			
	10% MCWB	4.8	6.2	8.0	11.0	14.5	17.2	18.2	18.1	15.6	12.8	10.2	6.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	9.0	10.3	11.8	14.7	18.1	20.3	21.1	21.2	18.7	16.5	13.8	11.4			
	0.4% MCDB	12.5	15.8	18.6	23.4	27.0	29.8	32.5	32.1	28.4	23.5	17.5	15.1			
	2% WB	7.5	8.7	10.4	13.6	16.8	19.4	20.2	20.3	17.7	15.4	12.6	9.8			
	2% MCDB	10.8	13.1	17.1	20.8	25.2	28.6	30.3	30.8	26.7	21.6	16.0	11.9			
	5% WB	6.3	7.5	9.5	12.6	15.9	18.6	19.5	19.5	17.1	14.5	11.7	8.5			
	5% MCDB	9.0	11.5	15.5	19.0	23.5	27.5	29.3	29.5	25.3	19.5	15.1	10.8			
	10% WB	5.1	6.5	8.5	11.7	15.1	17.9	18.9	18.7	16.4	13.7	10.7	7.1			
	10% MCDB	7.0	9.9	13.8	17.2	21.7	26.3	28.4	28.0	23.5	18.3	13.6	9.1			
(35) Mean Daily Temperature Range	MDBR	7.2	8.9	10.2	10.6	11.8	13.4	14.4	14.7	12.4	11.3	8.7	6.7			
	5% DB MCDBR	10.0	13.0	14.9	14.9	15.7	16.9	17.5	17.4	16.9	15.7	12.9	9.3			
	5% WB MCWBR	6.0	7.1	7.0	6.3	5.8	5.3	5.0	5.0	5.5	6.6	6.9	5.5			
	5% WB MCDBR	9.2	11.5	13.0	13.0	14.1	15.3	15.6	15.5	13.8	12.4	9.8	7.3			
(40) Clear-Sky Solar Irradiance	taub	0.329	0.359	0.408	0.470	0.463	0.438	0.442	0.457	0.442	0.398	0.359	0.331			
	taud	2.347	2.250	2.143	1.985	2.069	2.191	2.179	2.131	2.142	2.249	2.320	2.337			
	Ebn at Noon	813	838	833	804	821	841	833	807	789	782	766	775			
	Edh at Noon	86	110	137	173	163	144	145	148	136	109	89	81			
(44) All-Sky Solar Radiation	RadAvg	1.75	2.58	3.63	4.68	5.66	6.54	6.85	6.09	4.46	3.13	2.04	1.53			
	RadStd	0.25	0.33	0.44	0.53	0.38	0.49	0.39	0.38	0.38	0.34	0.21	0.27			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A

Nomenclature: See separate page